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November 10, 2025, 9:44 PM

• Search:

CAS Lexicon Search:

Paal-Knorr reaction - Preferred Concept

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Document Type:	Journal	

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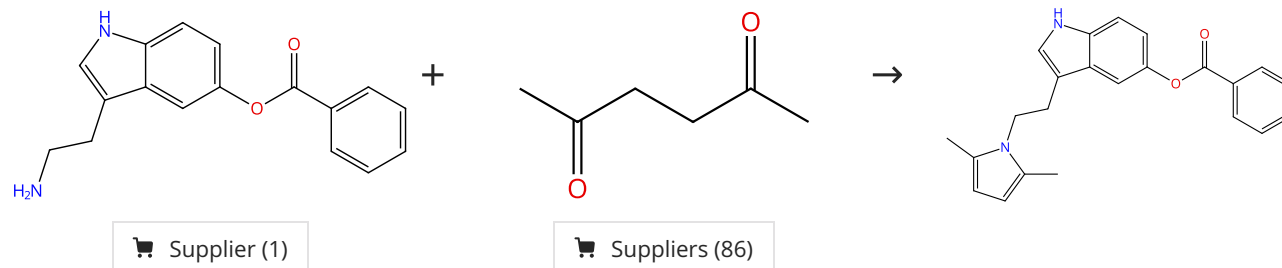
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-11293564

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 15 min, rt

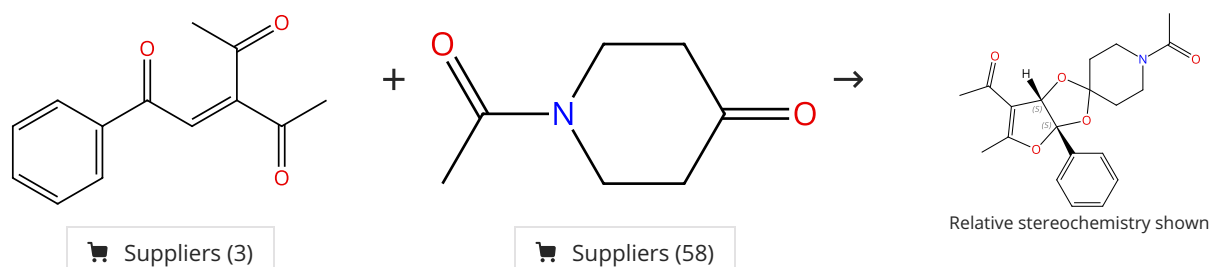
Experimental Protocols

By: Cho, Hyejin; et al

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-301-CAS-18526268

Steps: 1 Yield: 100%

Reaction of electron-deficient 3-acetyl-1-arylpent-2-ene-1,4-diones as a building block of hetero cycles

1.1 Reagents: Boron trifluoride etherate

Solvents: Dichloromethane; 0 °C; 2.5 h, -20 °C

By: Kawabata, Shun; et al

Heterocycles (2017), 94(8), 1479-1505.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-574-CAS-18526279

Steps: 1 Yield: 100%

Reaction of electron-deficient 3-acetyl-1-arylpent-2-ene-1,4-diones as a building block of hetero cycles

1.1 Reagents: Ammonium acetate

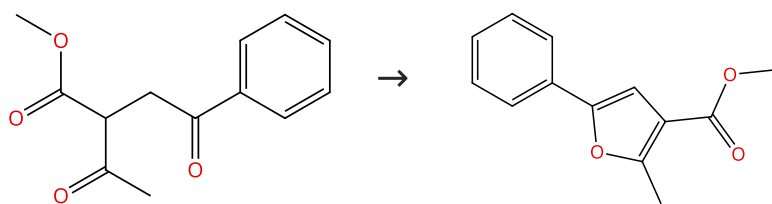
Solvents: Acetic acid; 45 min, 110 °C

By: Kawabata, Shun; et al

Heterocycles (2017), 94(8), 1479-1505.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (9)

31-614-CAS-25078793

Steps: 1 Yield: 100%

Infrared heat aided grinding synthesis of furan derivatives

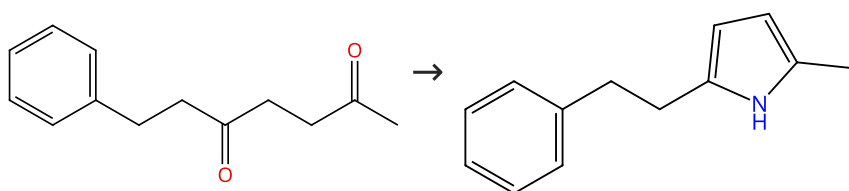
1.1 Catalysts: Silica, Sulfuric acid; 1 min, rt; 8 min, 80 °C

By: Li, Jinghua; et al

Zhejiang Gongye Daxue Xuebao (2016), 44(3), 351-354.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (19)

Suppliers (4)

31-574-CAS-8204691

Steps: 1 Yield: 100%

Flow synthesis using gaseous ammonia in a Teflon A F-2400 tube-in-tube reactor: Paal-Knorr pyrrole formation and gas concentration measurement by inline flow titration

1.1 Reagents: Ammonia

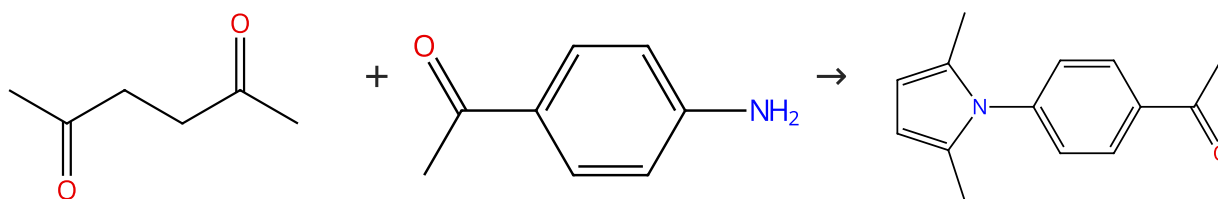
Solvents: Methanol; 3.5 bar, 0 °C; 6 bar, 110 °C

By: Cranwell, Philippa B.; et al

Organic & Biomolecular Chemistry (2012), 10(30), 5774-5779.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (102)

Suppliers (38)

31-309-CAS-22797406

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 90 min, 60 °C

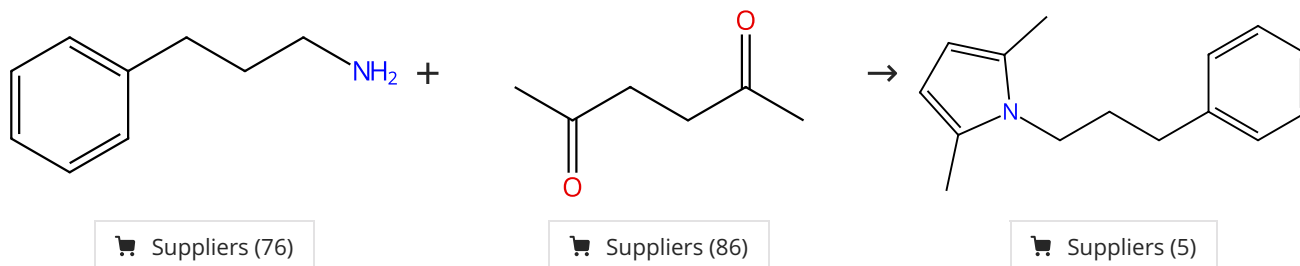
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-802573

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 5 min, rt

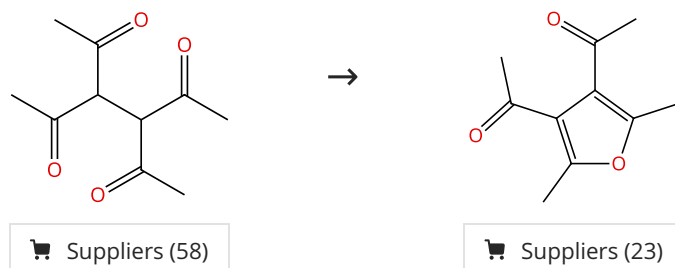
By: Cho, Hyejin; et al

Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-25078794

Steps: 1 Yield: 100%

Infrared heat aided grinding synthesis of furan derivatives

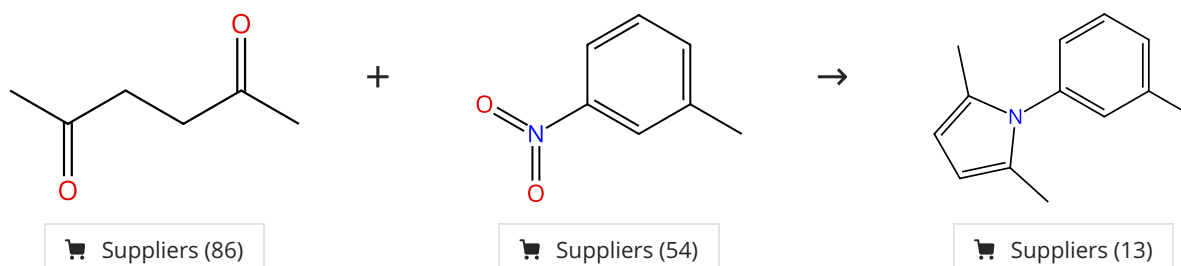
1.1 Catalysts: Silica, Sulfuric acid; 1 min, rt; 8 min, 80 °C

By: Li, Jinghua; et al

Zhejiang Gongye Daxue Xuebao (2016), 44(3), 351-354.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-17692505

Steps: 1 Yield: 100%

One-pot synthesis of N-substituted pyrroles from nitro compounds and 2,5-hexadione over a heterogeneous cobalt catalyst

1.1 Reagents: Formic acid

By: Gong, Zheng; et al

Catalysts: Cobalt (nitrogen-doped carbon-supported)

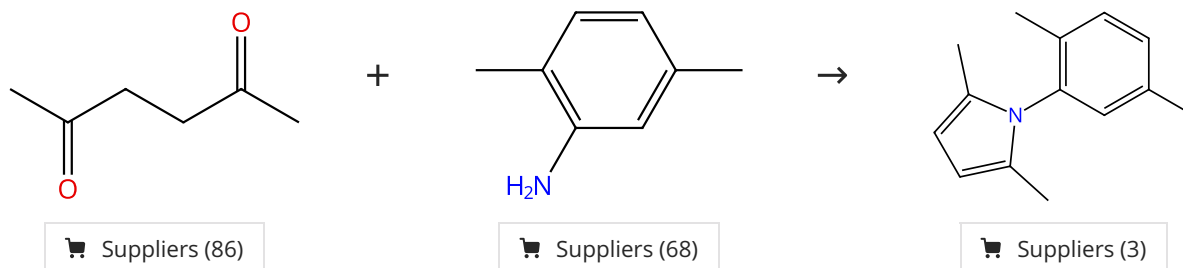
Solvents: Ethanol; 12 h, 110 °C

New Journal of Chemistry (2017), 41(19), 10613-10618.

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-22797399

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 30 min, 60 °C

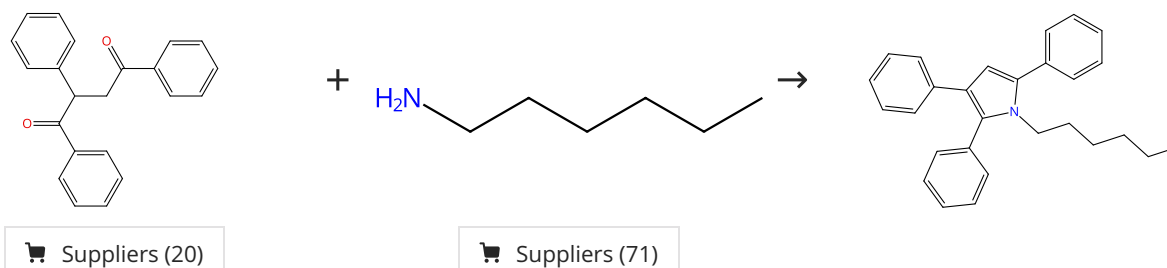
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-29437186

Steps: 1 Yield: 100%

A convenient approach for the synthesis of substituted pyrroles by using phosphoric acid as a catalyst and their photophysical properties

1.1 Catalysts: Phosphoric acid

Solvents: Water; 8 h, 140 °C; 140 °C → rt

By: Ibrahim, Yusuf Ajibola; et al

1.2 Reagents: Hydrochloric acid

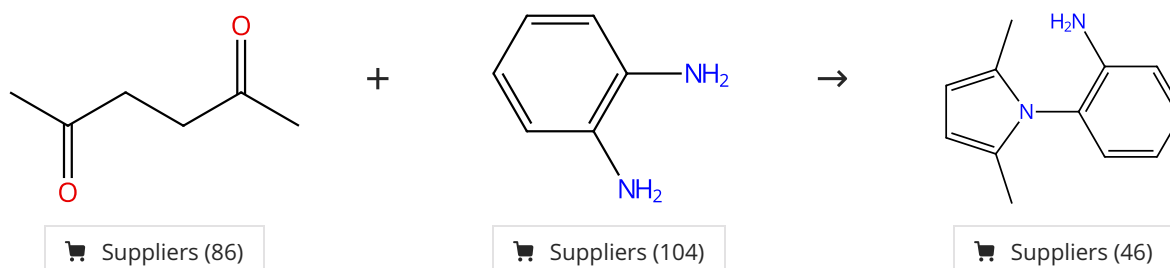
Solvents: Water; neutralized

Journal of Molecular Structure (2022), 1252, 132123.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-16923154

Steps: 1 Yield: 100%

Dendritic Amine on Mesoporous Silica: First Organo Base Catalyst for Paal Knorr Reaction under Solvent Free Condition, A green approach

1.1 Catalysts: 2085771-29-3; 2 h, rt

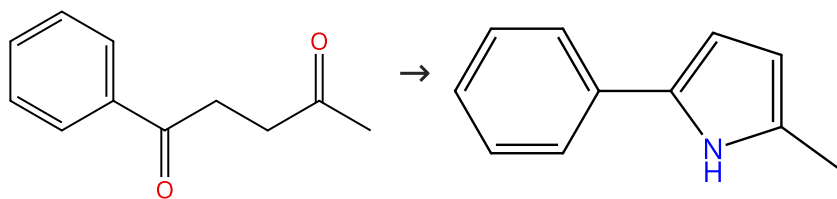
Experimental Protocols

By: Jisha, K. A.; et al

Catalysis Letters (2017), 147(4), 964-975.

Scheme 13 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (42)

Suppliers (17)

31-574-CAS-3800323

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

By: Cho, Hyejin; et al

Green Chemistry (2015), 17(2), 1088-1099.

1.1 Reagents: Ammonium hydroxide
Solvents: Water; 5 h, rt

Experimental Protocols

31-574-CAS-13074844

Steps: 1 Yield: 100%

Flow synthesis using gaseous ammonia in a Teflon A F-2400 tube-in-tube reactor: Paal-Knorr pyrrole formation and gas concentration measurement by inline flow titration

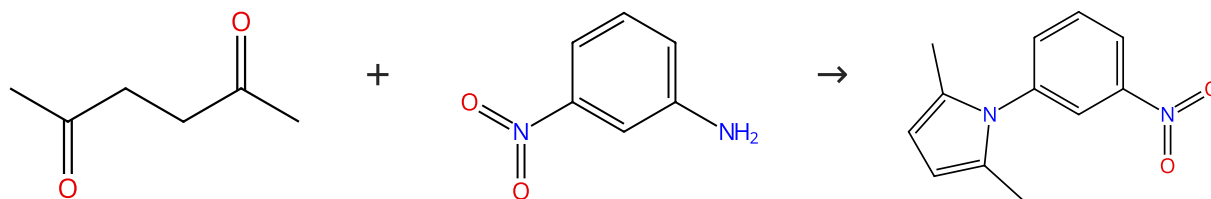
By: Cranwell, Philippa B.; et al

Organic & Biomolecular Chemistry (2012), 10(30), 5774-5779.

1.1 Reagents: Ammonia
Solvents: Methanol; 3.5 bar, 0 °C; 6 bar, 110 °C

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (60)

Suppliers (29)

31-309-CAS-22797409

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

By: Mohsenzadeh, Farshid; et al

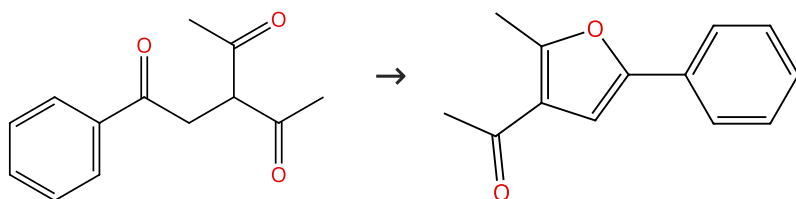
Research on Chemical Intermediates (2020), 46(12), 5255-5262.

1.1 Catalysts: Oxalic acid
Solvents: Ethanol; 60 min, 60 °C

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (7)

31-614-CAS-25078803

Steps: 1 Yield: 100%

Infrared heat aided grinding synthesis of furan derivatives

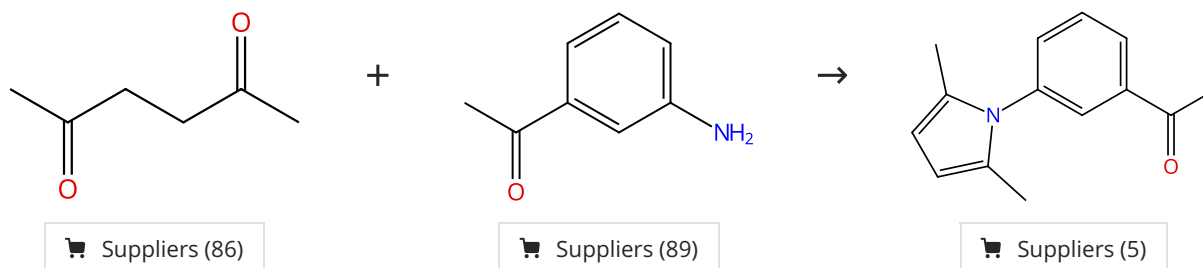
By: Li, Jinghua; et al

Zhejiang Gongye Daxue Xuebao (2016), 44(3), 351-354.

1.1 Catalysts: Silica, Sulfuric acid; 1 min, rt; 6 min, rt

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-22797407

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid
Solvents: Ethanol; 30 min, 60 °C

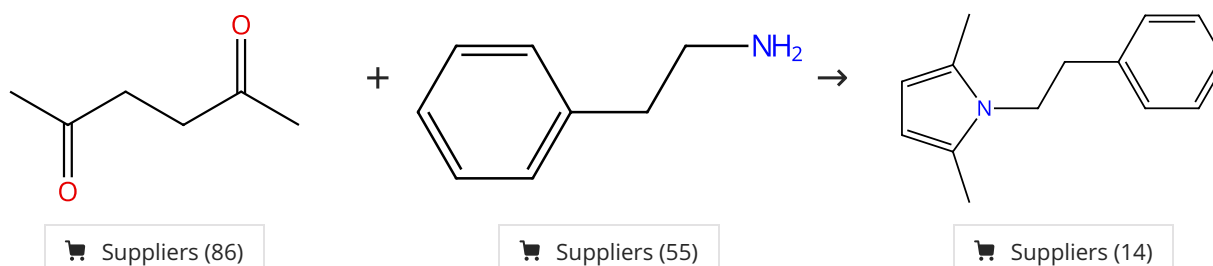
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-5325967

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 7 min, rt

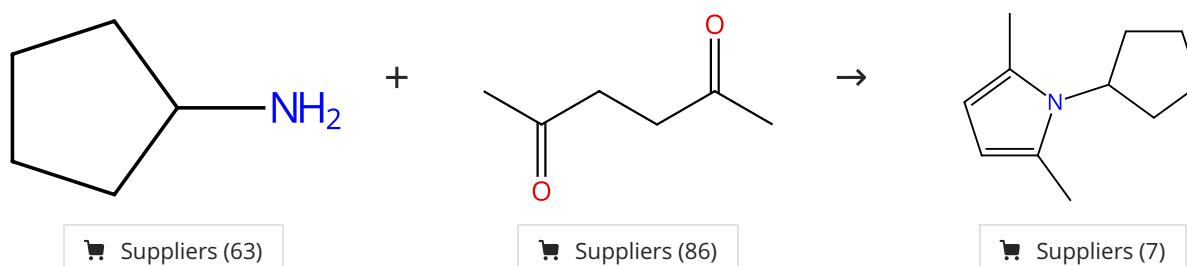
By: Cho, Hyejin; et al

Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 18 (2 Reactions)

Steps: 1 Yield: 99-100%



31-309-CAS-8609107

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 24 h, rt

By: Cho, Hyejin; et al

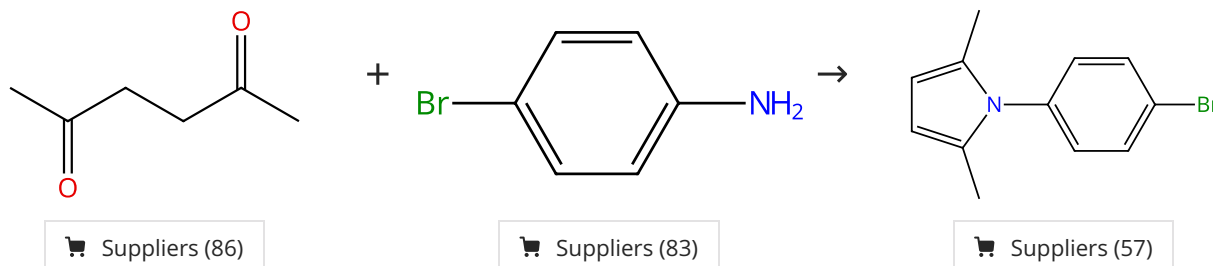
Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

31-309-CAS-19108336	Steps: 1 Yield: 99%	Novel and highly efficient preparation of pyrroles using supported ionic liquid ILCF₃SO₃@SiO₂ as a heterogeneous catalyst
1.1 Catalysts: Silica, 1 <i>H</i> -imidazolium, 1-(4-sulfobutyl)-3-[3-(triethoxysilyl)propyl]-, 1,1,1-trifluoromethanesulfonate (1:1) (silica-supported) Solvents: Ethanol; 5 min, rt		By: Liu, Yang; et al
Experimental Protocols		Journal of the Iranian Chemical Society (2018), 15(5), 1033-1040.

Scheme 19 (2 Reactions)

Steps: 1 Yield: 100%

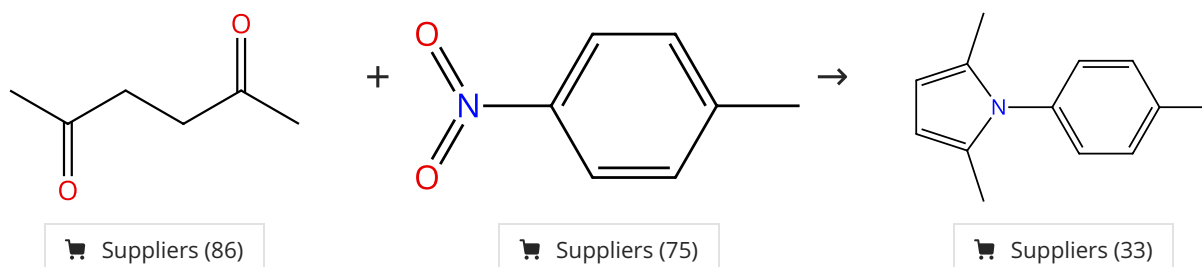


31-309-CAS-22797400	Steps: 1 Yield: 100%	Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction
1.1 Catalysts: Oxalic acid Solvents: Ethanol; 30 min, 60 °C		By: Mohsenzadeh, Farshid; et al
Experimental Protocols		Research on Chemical Intermediates (2020), 46(12), 5255-5262.

31-309-CAS-3971775	Steps: 1 Yield: 100%	L-Tryptophan-catalyzed Paal-Knorr pyrrole cyclocondensation: an efficient, clean and recyclable organocatalyst
1.1 Catalysts: L-Tryptophan; 1 h, 70 °C		By: Aghapoor, Kioumars; et al
Experimental Protocols		Research on Chemical Intermediates (2016), 42(2), 407-415.

Scheme 20 (1 Reaction)

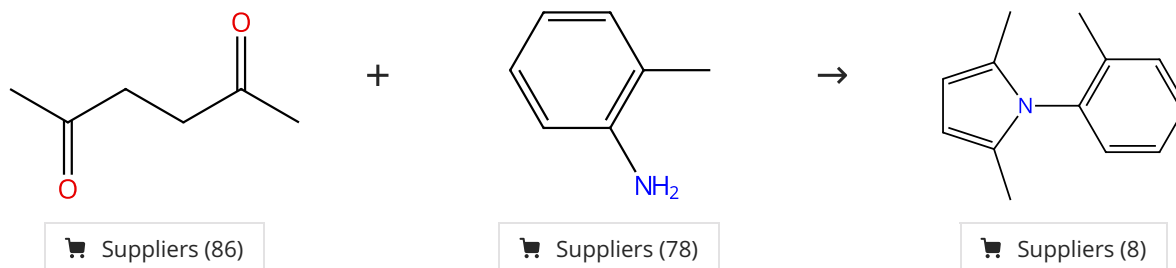
Steps: 1 Yield: 100%



31-309-CAS-17692504	Steps: 1 Yield: 100%	One-pot synthesis of N-substituted pyrroles from nitro compounds and 2,5-hexadione over a heterogeneous cobalt catalyst
1.1 Reagents: Formic acid Catalysts: Cobalt (nitrogen-doped carbon-supported) Solvents: Ethanol; 12 h, 110 °C		By: Gong, Zheng; et al
Experimental Protocols		New Journal of Chemistry (2017), 41(19), 10613-10618.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-22797397

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid
Solvents: Ethanol; 30 min, 60 °C

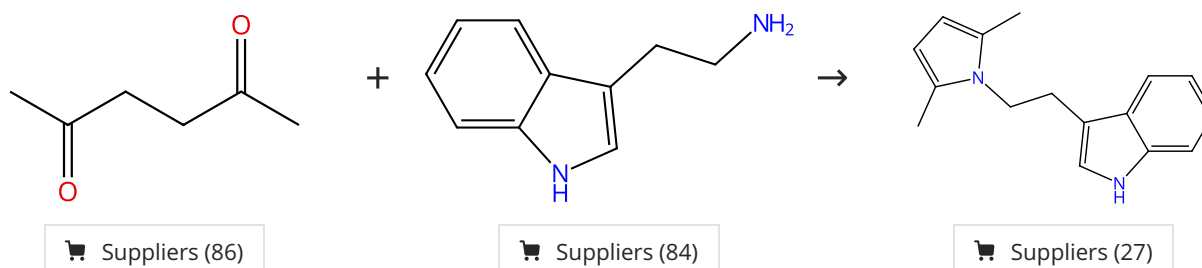
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-6429124

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 5 min, rt

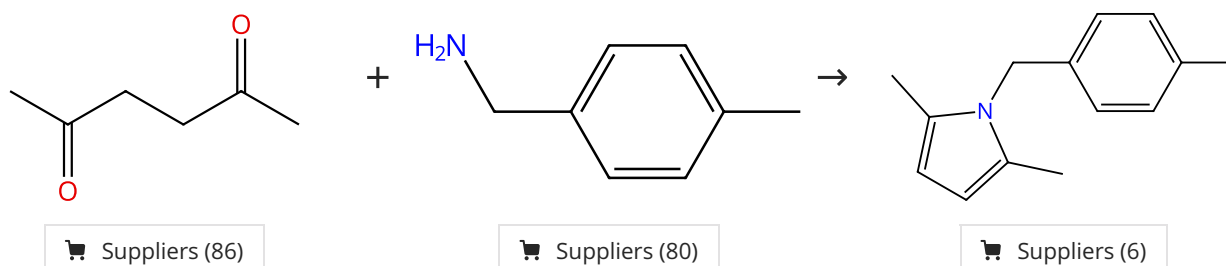
By: Cho, Hyejin; et al

Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-736532

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 10 min, rt

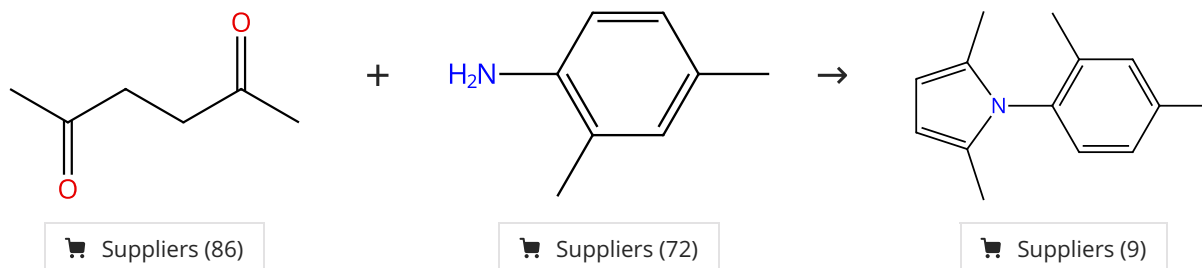
By: Cho, Hyejin; et al

Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



31-309-CAS-22797398

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 30 min, 60 °C

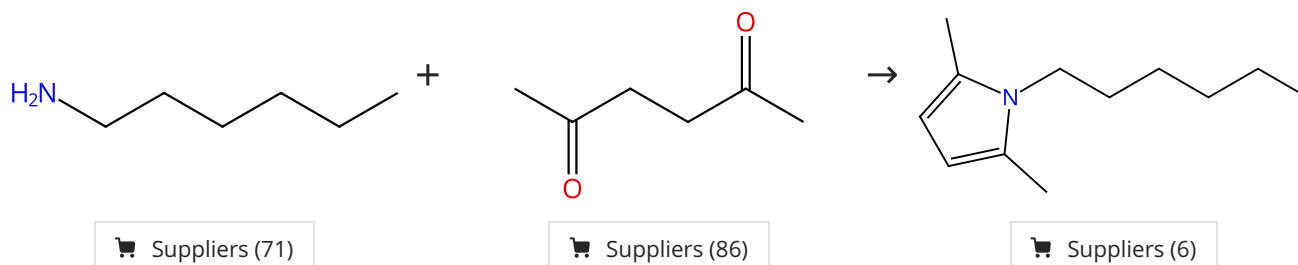
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 25 (2 Reactions)

Steps: 1 Yield: 100%



31-309-CAS-22797413

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 30 min, 60 °C

By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

31-309-CAS-3974168

Steps: 1 Yield: 100%

The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles

1.1 7 min, rt

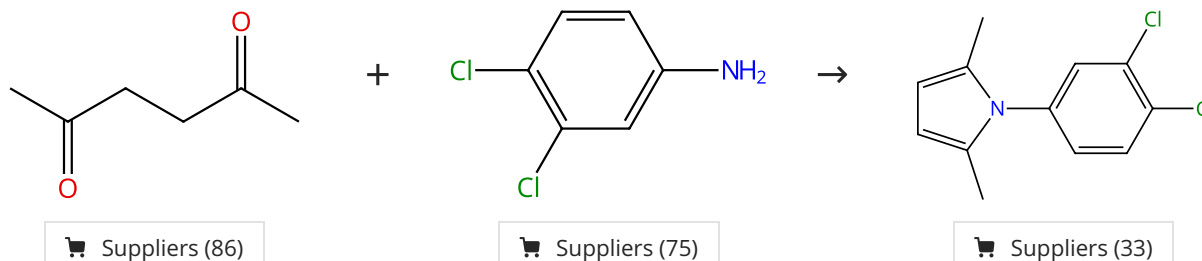
By: Cho, Hyejin; et al

Experimental Protocols

Green Chemistry (2015), 17(2), 1088-1099.

Scheme 26 (1 Reaction)

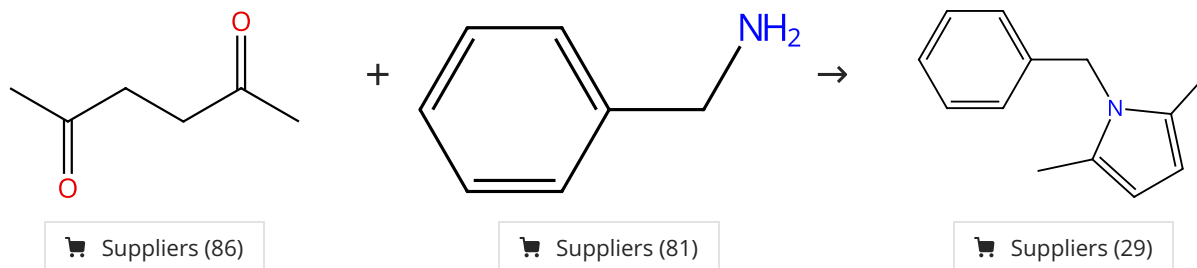
Steps: 1 Yield: 100%



31-309-CAS-22797403	Steps: 1 Yield: 100%	Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction
1.1 Catalysts: Oxalic acid Solvents: Ethanol; 30 min, 60 °C		By: Mohsenzadeh, Farshid; et al
Experimental Protocols		Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 27 (4 Reactions)

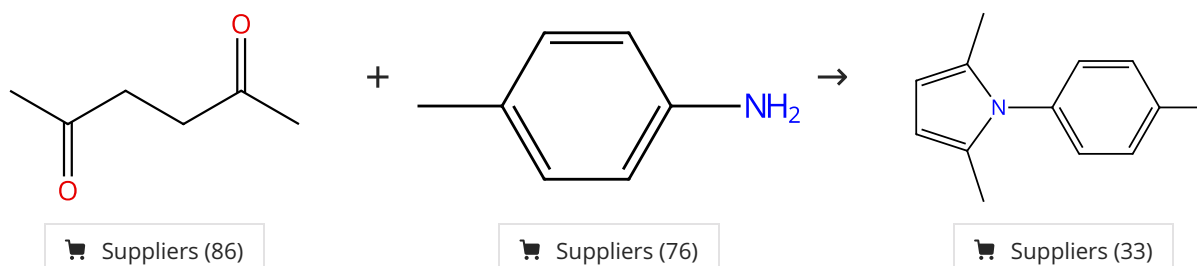
Steps: 1 Yield: 99-100%



31-614-CAS-37331192	Steps: 1 Yield: 100%	Ascorbic acid as a multifunctional hydrogen bonding catalyst for Paal-Knorr synthesis of N-substituted mono- and bis-pyrroles: experimental and theoretical aspects
1.1 Catalysts: Ascorbic acid; 1 h, 70 °C		By: Ghaemi, Amin; et al
Experimental Protocols		Research on Chemical Intermediates (2023), 49(9), 4087-4102.
31-309-CAS-23445263	Steps: 1 Yield: 99%	Hexafluoroisopropanol as solvent and promoter in the Paal-Knorr synthesis of N-substituted diaryl pyrroles
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 20 °C; 5 min, 20 °C		By: Schirmacher, Robert H. E.; et al
Experimental Protocols		Tetrahedron (2021), 83, 131985.
31-309-CAS-9642306	Steps: 1 Yield: 99%	A novel enzyme-catalyzed synthesis of N-substituted pyrrole derivatives
1.1 Catalysts: α-Amylase Solvents: Methanol; 2 h, 50 °C		By: Zheng, Hui; et al
Experimental Protocols		Molecular Diversity (2013), 17(2), 245-250.
31-309-CAS-11464800	Steps: 1 Yield: 99%	Paal-Knorr pyrrole synthesis using recyclable amberlite I R 120 acidic resin, a green approach
1.1 Catalysts: Amberlite IR 120 (acidic resin bound); 16 min, rt		By: Devi, Aarti; et al
Experimental Protocols		Synthetic Communications (2012), 42(10), 1480-1488.

Scheme 28 (2 Reactions)

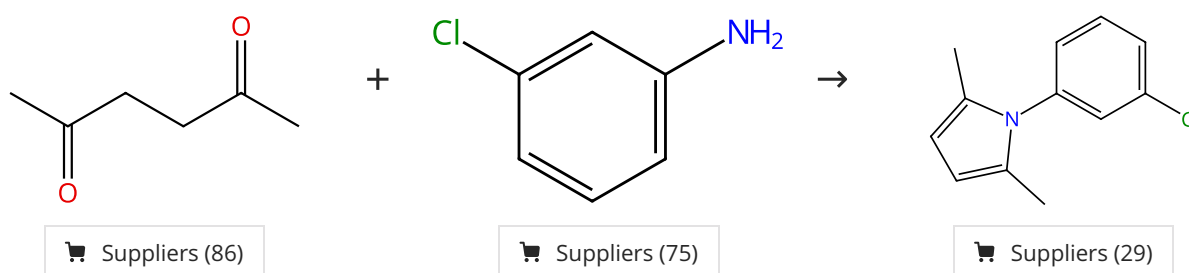
Steps: 1 Yield: 100%



31-614-CAS-37331172 Steps: 1 Yield: 100% 1.1 Catalysts: Ascorbic acid; 1 h, 70 °C Experimental Protocols	Ascorbic acid as a multifunctional hydrogen bonding catalyst for Paal-Knorr synthesis of N-substituted mono- and bis-pyrroles: experimental and theoretical aspects By: Ghaemi, Amin; et al Research on Chemical Intermediates (2023), 49(9), 4087-4102.
31-309-CAS-22797395 Steps: 1 Yield: 100% 1.1 Catalysts: Oxalic acid Solvents: Ethanol; 30 min, 60 °C Experimental Protocols	Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction By: Mohsenzadeh, Farshid; et al Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 29 (1 Reaction)

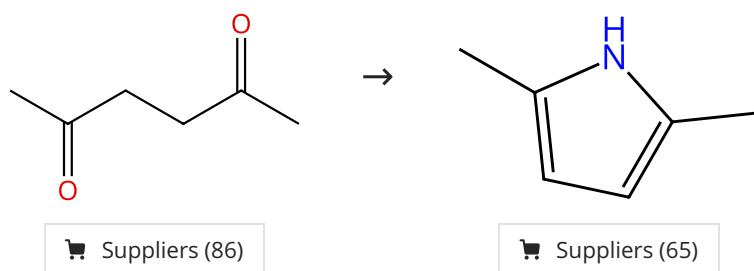
Steps: 1 Yield: 100%



31-309-CAS-22797402 Steps: 1 Yield: 100% 1.1 Catalysts: Oxalic acid Solvents: Ethanol; 30 min, 60 °C Experimental Protocols	Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction By: Mohsenzadeh, Farshid; et al Research on Chemical Intermediates (2020), 46(12), 5255-5262.
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Scheme 30 (2 Reactions)

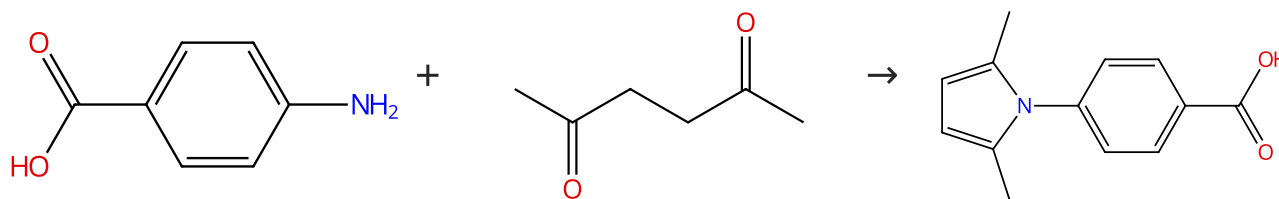
Steps: 1 Yield: 100%



31-574-CAS-619836 Steps: 1 Yield: 100% 1.1 Reagents: Ammonium, hydrate (1:1) Solvents: Water; 3 h, rt Experimental Protocols	The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles By: Cho, Hyejin; et al Green Chemistry (2015), 17(2), 1088-1099.
31-574-CAS-5215373 Steps: 1 Yield: 100% 1.1 Reagents: Ammonia Solvents: Methanol; 3.5 bar, 0 °C; 6 bar, 110 °C	Flow synthesis using gaseous ammonia in a Teflon A F-2400 tube-in-tube reactor: Paal-Knorr pyrrole formation and gas concentration measurement by inline flow titration By: Cranwell, Philippa B.; et al Organic & Biomolecular Chemistry (2012), 10(30), 5774-5779.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (129)

Suppliers (86)

Suppliers (66)

31-309-CAS-22797405

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 30 min, 60 °C

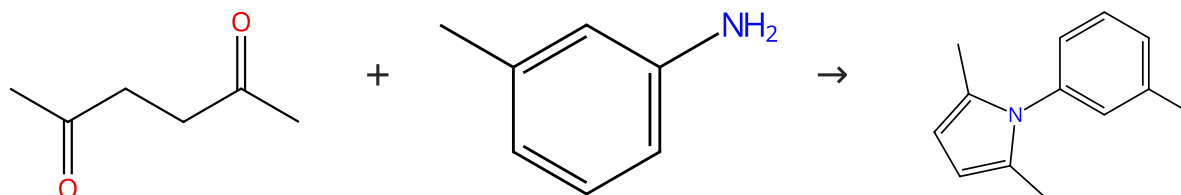
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (73)

Suppliers (13)

31-309-CAS-22797396

Steps: 1 Yield: 100%

Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction

1.1 Catalysts: Oxalic acid

Solvents: Ethanol; 30 min, 60 °C

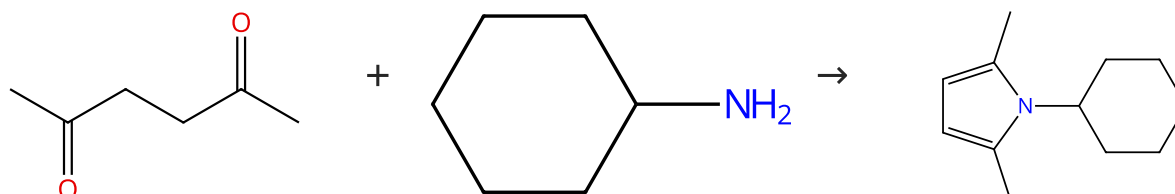
By: Mohsenzadeh, Farshid; et al

Experimental Protocols

Research on Chemical Intermediates (2020), 46(12), 5255-5262.

Scheme 33 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (85)

Suppliers (19)

31-614-CAS-37331186

Steps: 1 Yield: 100%

Ascorbic acid as a multifunctional hydrogen bonding catalyst for Paal-Knorr synthesis of N-substituted mono- and bis-pyrroles: experimental and theoretical aspects

1.1 Catalysts: Ascorbic acid; 1 h, 70 °C

Experimental Protocols

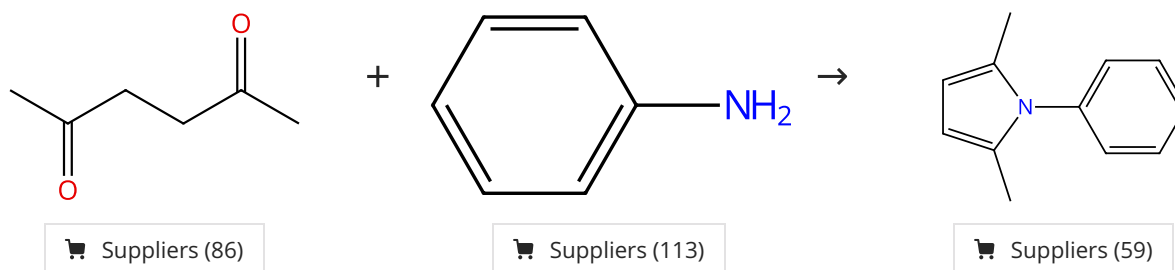
By: Ghaemi, Amin; et al

Research on Chemical Intermediates (2023), 49(9), 4087-4102.

31-309-CAS-13523483	Steps: 1 Yield: 100%	The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles By: Cho, Hyejin; et al Green Chemistry (2015), 17(2), 1088-1099.
1.1 24 h, rt		
Experimental Protocols		

Scheme 34 (3 Reactions)

Steps: 1 Yield: 100%



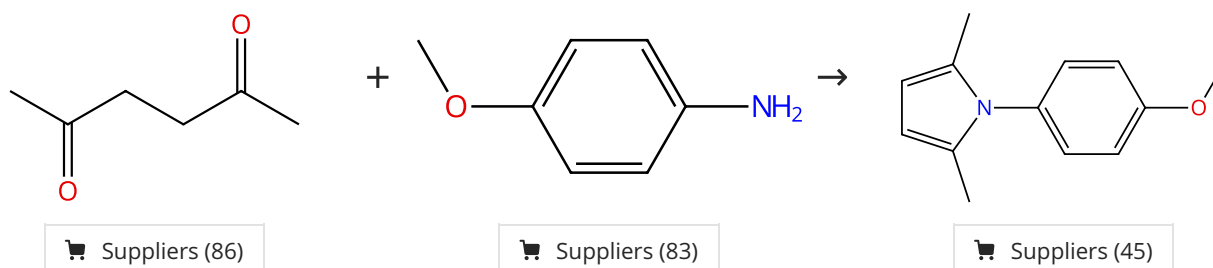
31-614-CAS-37331173	Steps: 1 Yield: 100%	Ascorbic acid as a multifunctional hydrogen bonding catalyst for Paal-Knorr synthesis of N-substituted mono- and bis-pyrroles: experimental and theoretical aspects By: Ghaemi, Amin; et al Research on Chemical Intermediates (2023), 49(9), 4087-4102.
1.1 Catalysts: Ascorbic acid; 1 h, 70 °C		
Experimental Protocols		

31-309-CAS-22797394	Steps: 1 Yield: 100%	Naturally occurring organic acids for organocatalytic synthesis of pyrroles via Paal-Knorr reaction By: Mohsenzadeh, Farshid; et al Research on Chemical Intermediates (2020), 46(12), 5255-5262.
1.1 Catalysts: Oxalic acid Solvents: Ethanol; 30 min, 60 °C		
Experimental Protocols		

31-309-CAS-3877943	Steps: 1 Yield: 100%	Paal-Knorr pyrrole synthesis in water By: Akbaslar, Dilek; et al Synthetic Communications (2014), 44(9), 1323-1332.
1.1 Catalysts: Bismuth trinitrate pentahydrate; 5 min, 90 °C		
Experimental Protocols		

Scheme 35 (1 Reaction)

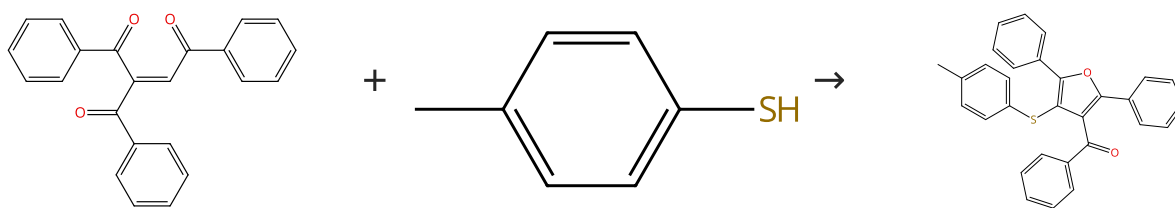
Steps: 1 Yield: 100%



31-309-CAS-2819954	Steps: 1 Yield: 100%	The Paal-Knorr reaction revisited. A catalyst and solvent-free synthesis of underivatized and N-substituted pyrroles By: Cho, Hyejin; et al Green Chemistry (2015), 17(2), 1088-1099.
1.1 4 h, rt		
Experimental Protocols		

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (56)

31-571-CAS-19542587

Steps: 1 Yield: 99%

A one-pot metal-free protocol for the synthesis of chalcogenated furans from 1,4-enediones and thiols

By: Rajeshkumar, Venkatachalam; et al

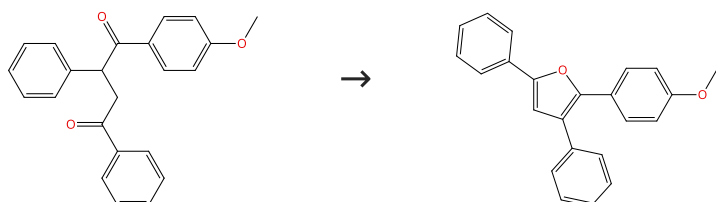
Organic & Biomolecular Chemistry (2019), 17(4), 982-991.

1.1 **Catalysts:** *p*-Toluenesulfonic acid
Solvents: 1,2-Dichloroethane; 5 h, 100 °C

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

31-614-CAS-24395300

Steps: 1 Yield: 99%

Synthesis of Arylfurans by Organic-Solvent-Free Method Using Phosphoric Acid as a Solvent and Catalyst

By: Ajibola, Ibrahim Yusuf; et al

ChemistrySelect (2021), 6(36), 9559-9564.

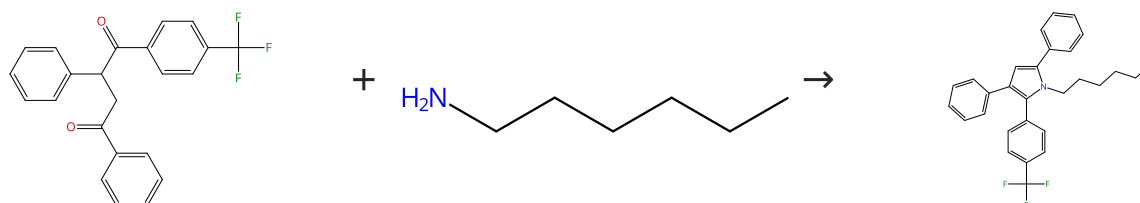
1.1 **Catalysts:** Phosphoric acid; 3 h, 140 °C

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (71)

31-614-CAS-29437199

Steps: 1 Yield: 99%

A convenient approach for the synthesis of substituted pyrroles by using phosphoric acid as a catalyst and their photophysical properties

By: Ibrahim, Yusuf Ajibola; et al

Journal of Molecular Structure (2022), 1252, 132123.

1.1 **Catalysts:** Phosphoric acid
Solvents: Water; 8 h, 140 °C; 140 °C → rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized

Experimental Protocols